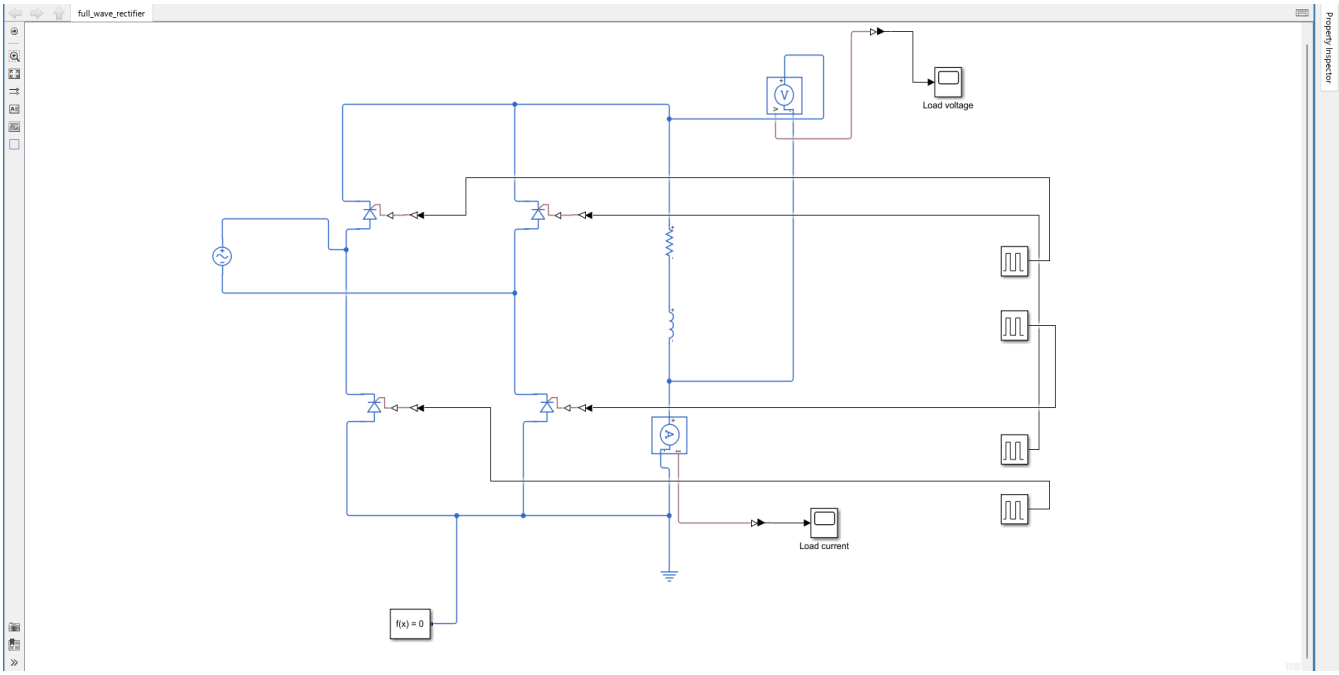
Transient, Steady-State, and Firing Angle (alpha) Analysis of a Full-Wave Controlled Thyristor Bridge Rectifier

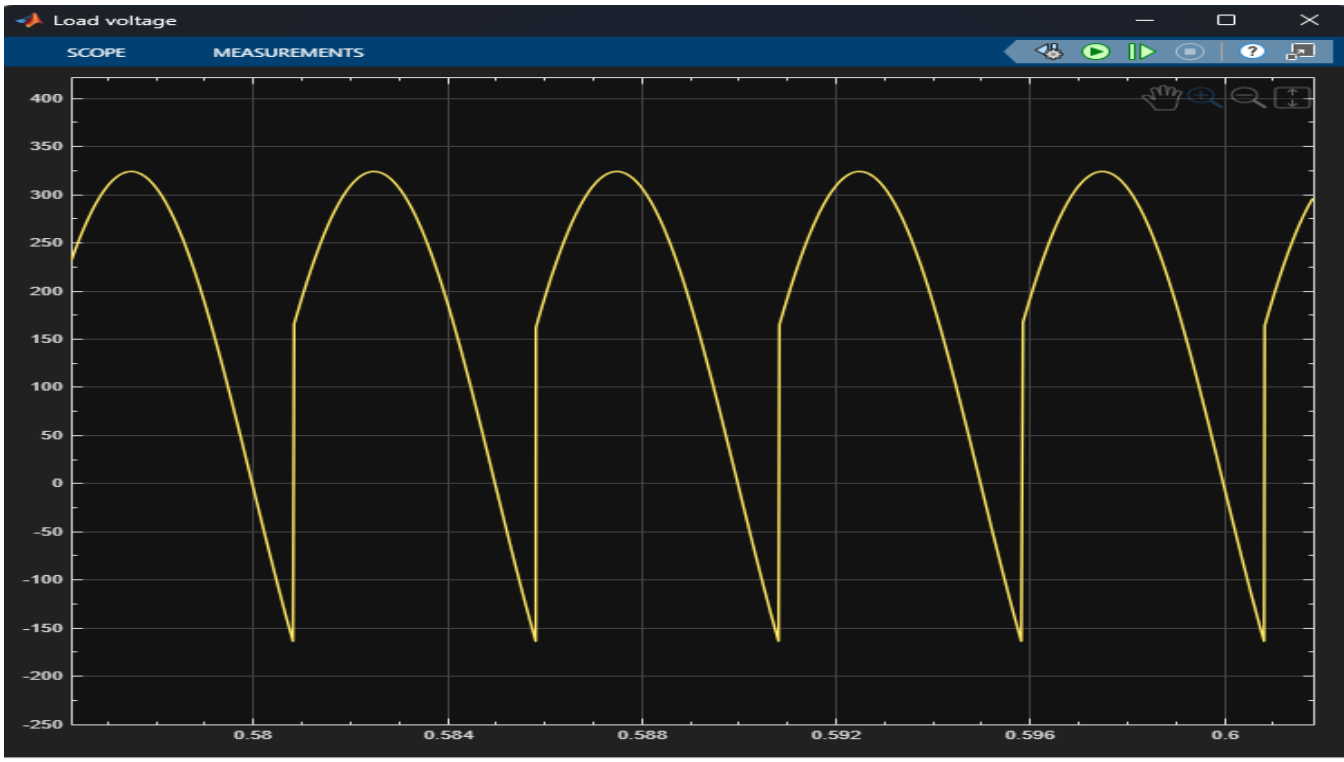
Author: Vishnuvarthan Balasubramanian

Executive Summary:

Investigation of single-phase full-wave controlled thyristor bridge rectifier performance supplying an inductive R-L load (R = 10 Ω, L = 20 mH) across firing angles alpha = 30, 60, and 90 deg (Vs = 230 Vrms, f = 100 Hz). Simulink bridge schematics and scope waveforms are presented on Pages 1–6, followed by complete first-principles KVL differential equation derivations (I_total = I_ss + I_dyn) and hand-to-Simulink voltage/current verification on Pages 7–12.



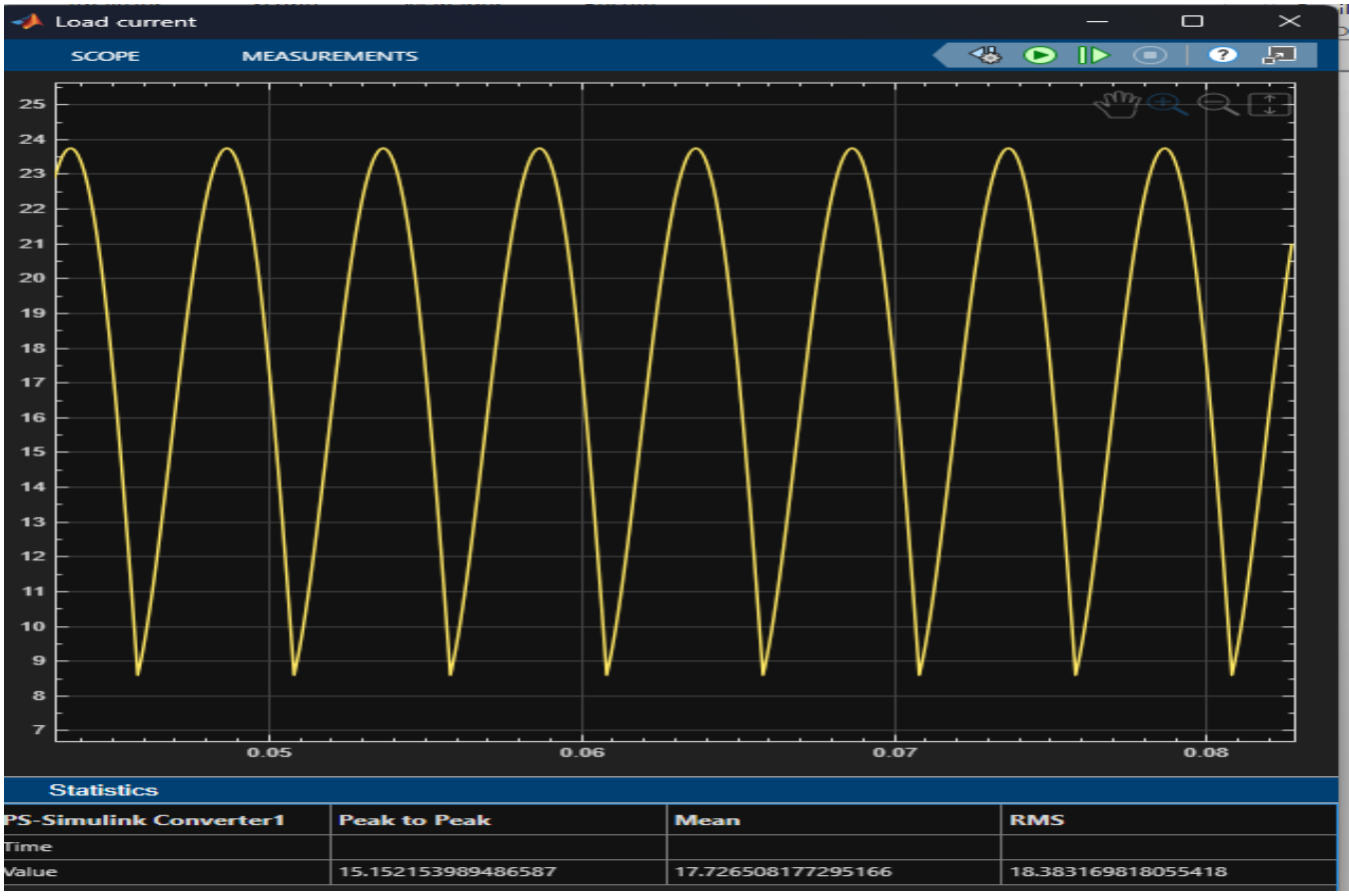
Load voltage (30 degree):



Statistics

PS-Simulink Converter	Peak to Peak	Mean	RMS
Time (ms)			
Value	487.8075682884182	179.68421465936146	229.6356492467469

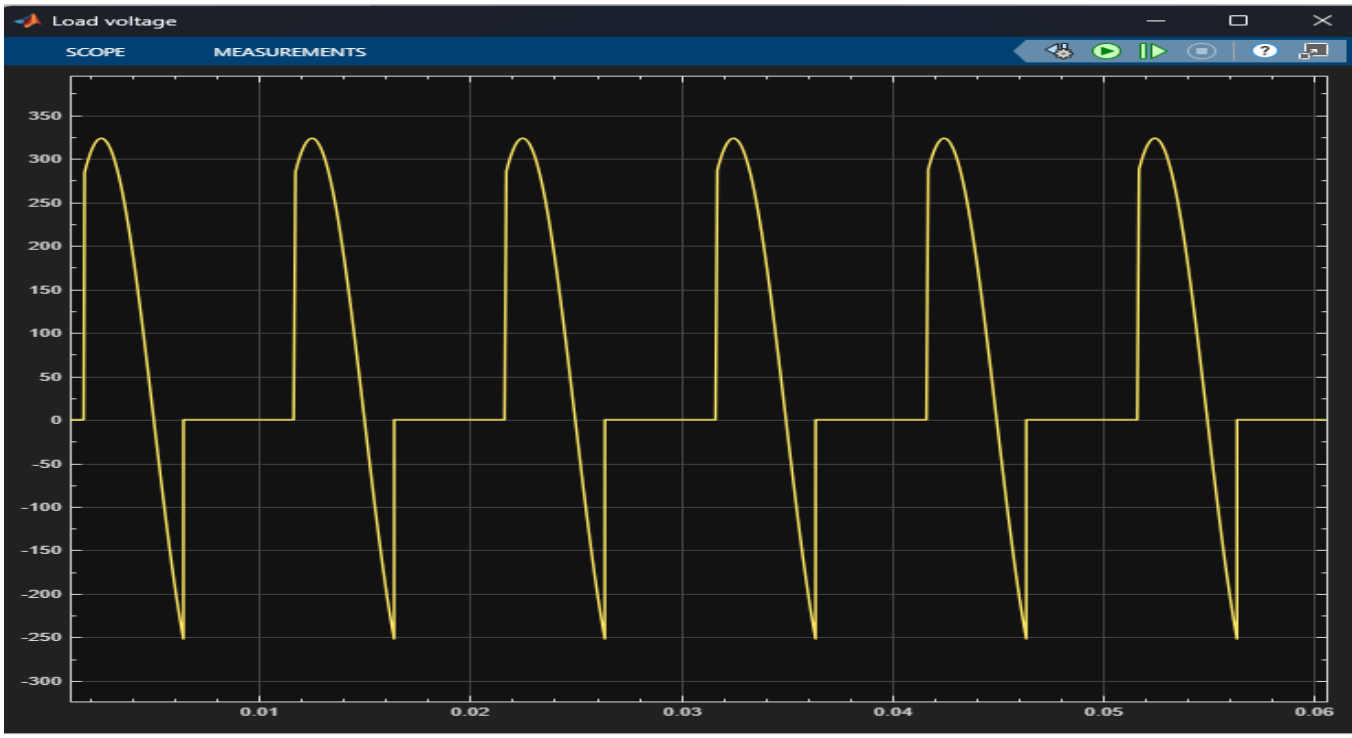
Load current (30 degree):



Statistics

PS-Simulink Converter1	Peak to Peak	Mean	RMS
Time			
Value	15.152153989486587	17.726508177295166	18.383169818055418

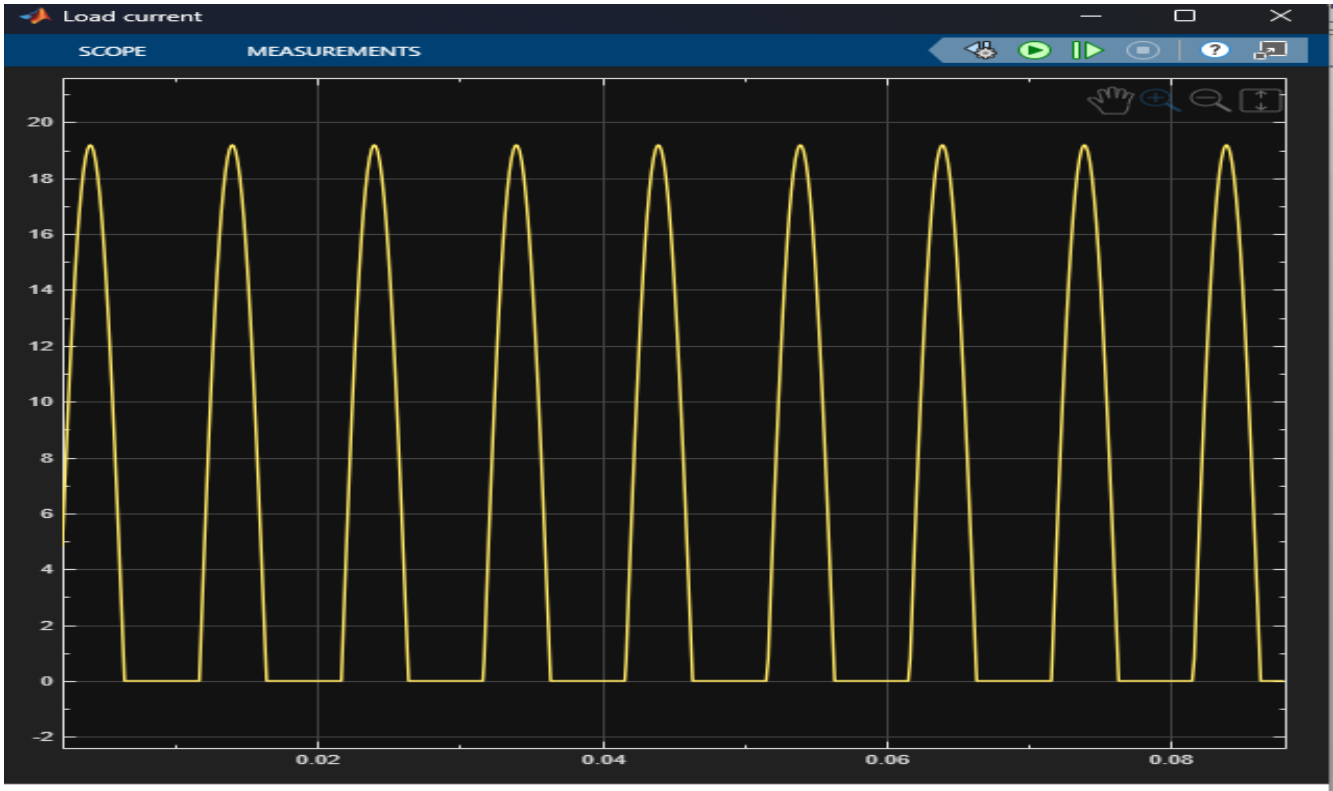
Load voltage (60 degree):



Statistics

PS-Simulink Converter	Peak to Peak	Mean	RMS
Time (ms)			
Value	575.3945704706179	57.61818690752368	156.25173110132158

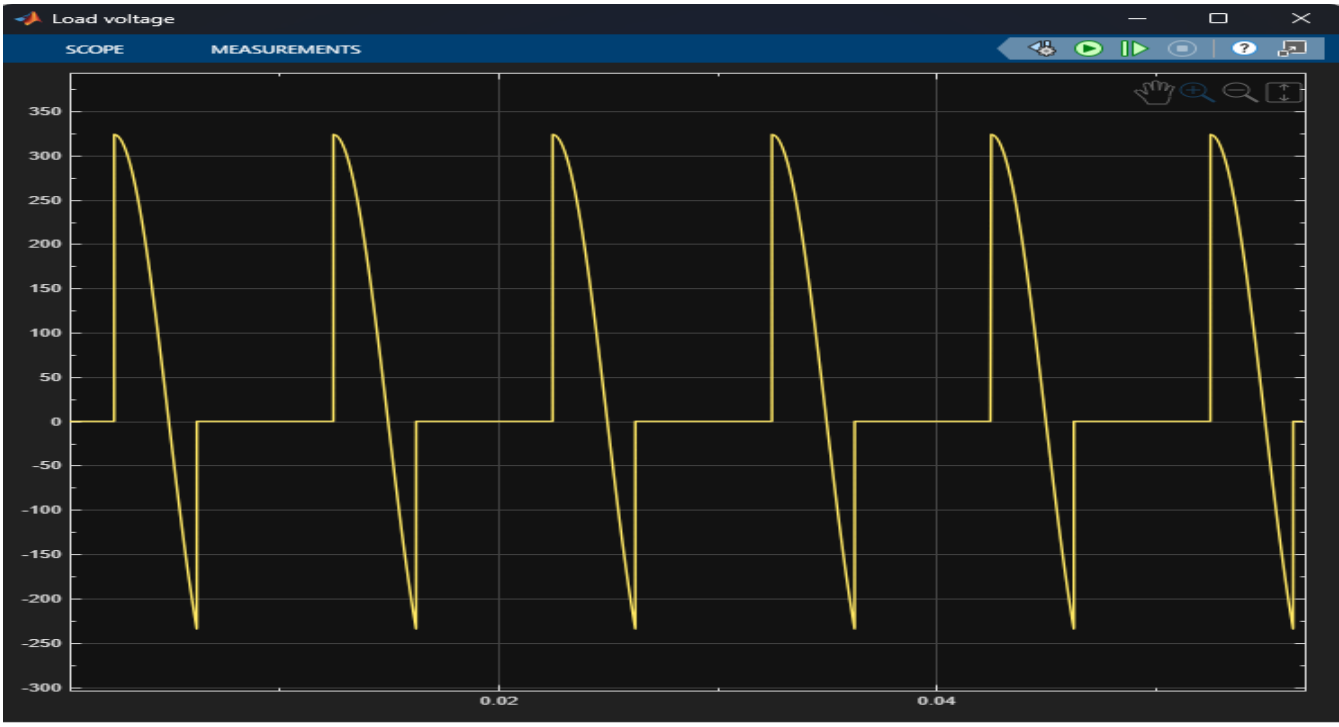
Load current (60 degree):



Statistics

PS-Simulink Converter1	Peak to Peak	Mean	RMS
Time (ms)			
Value	19.18944498563124	6.02420603464414 75	9.547164877229806

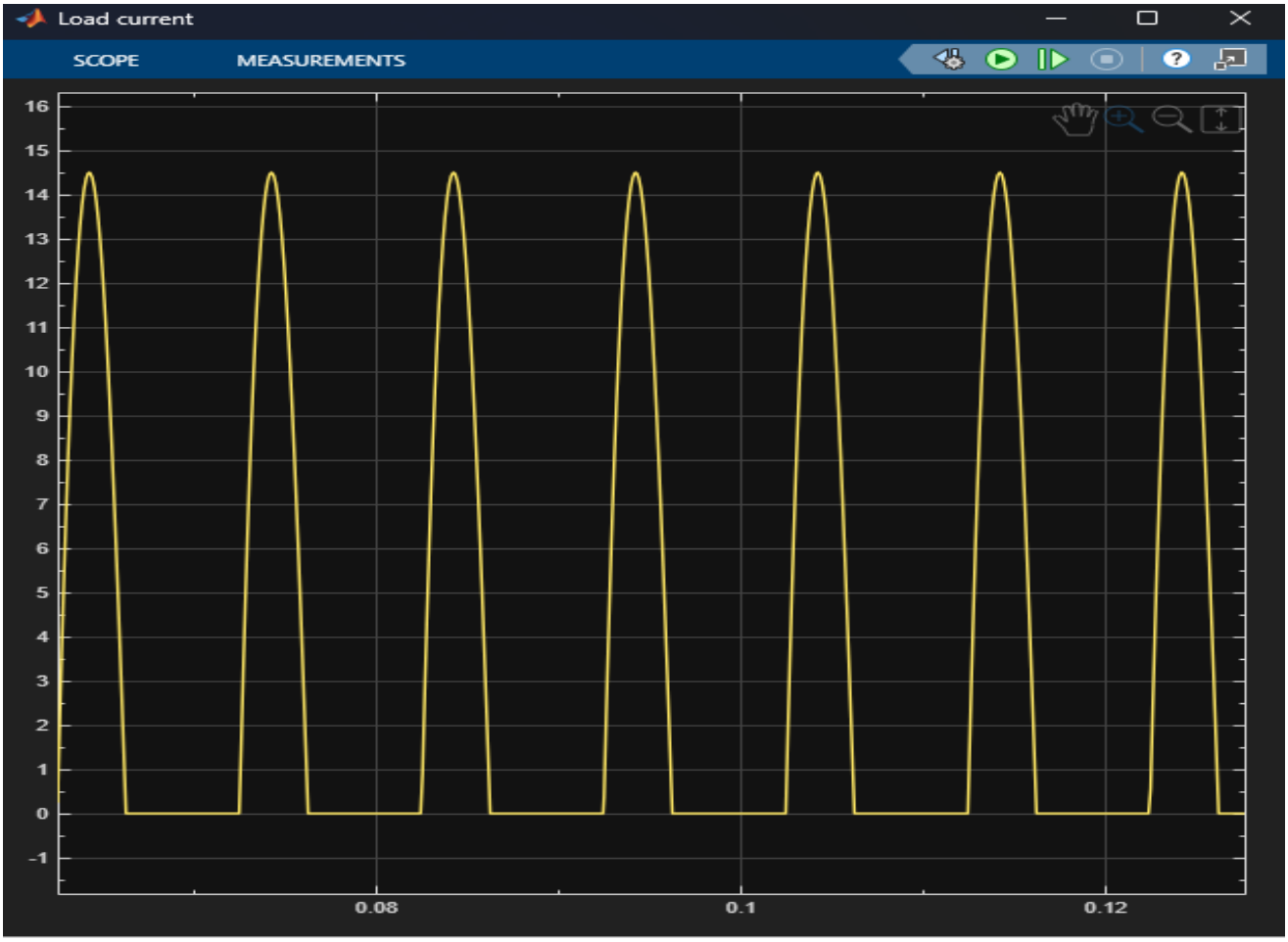
Load voltage (90 degree):



Statistics

PS-Simulink Converter	Peak to Peak	Mean	RMS
Time (ms)			
Value	557.9123521405071	37.17778748487836	128.88102599142752

Load current (90 degree):



Statistics

PS-Simulink Converter1	Peak to Peak	Mean	RMS
Time (ms)			
Value	14.494647063596528	3.788470926506576	6.595730911551006

Full wave Rectifier [Thyristor Controlled]. [Vishnuvardhan Ex-3]

① given: $V_s = 230$ rms, $F = 100$ Hz, $L = 20 \times 10^{-3}$ H, $R = 10$

$\alpha = 30°, 60°, 90°$.

1) $V_m = 230 \times \sqrt{2} = 325.27$ V

2) $\omega = 2\pi F = 2 \times \pi \times 100 = \boxed{200\pi = \omega}$

3) $X_L = \omega L = 200\pi \times 20 \times 10^{-3} \Rightarrow \boxed{4\pi = \omega L = X_L}$

4) $R = 10\,\Omega$

5) $Z = \sqrt{(10)^2 + (4\pi)^2}$ $\boxed{Z = 16.06\,\Omega}$

6) $\theta = \tan^{-1}\left(\frac{4\pi}{10}\right) \Rightarrow \boxed{\theta = 0.89863}$

Inductor so, $-\theta$.

$V_{source} = V_{drop\ in\ Resr} + V_{drop\ in\ Indr}$.

$325.27 \sin(200\pi t) = IR + L\frac{di}{dt}$.

$I_{total} = I_{ss} + I_{dy}$.

$$I_{ss} = \frac{325.27 \sin(200\pi t - 0.89863)}{16.06}$$

$$I_{ss} = 20.253 \sin(200\pi t - 0.89863)$$

I_{dyn} :

$$0 = I_{dyn} R + L \frac{d\, i_{dyn}}{dt}$$

$$-I_{dyn} R = L \frac{d\, i_{dyn}}{dt}$$

$$-dt\, R = L \frac{d\, I_{dyn}}{I_{dyn}}$$

$$-dt\, R/L = \frac{d\, I_{dyn}}{I_{dyn}}$$

$$\boxed{\int -R/L\, dt = \frac{1}{I_{dyn}} d\, I_{dyn}}$$

$$-R/L \int dt = \int \frac{1}{I_{dyn}} d\, I_{dyn}$$

$$-R/L\, t + c = \ln(I_{dyn})$$

$$e^{-R/L\, t + c} = I_{dyn}$$

$$x^a \times x^b = x^{a+b}$$

$$\boxed{e^{-R/L\, t} \times e^c = I_{dyn}}$$

$$I_{total} = I_{ss} + I_{dy}$$

$$\boxed{I_{total} = 20.253 \sin(200\pi t - 0.89863) + e^{-R/L\, t} \times e^c}$$

α depends on 2 Condus: $\theta = 51^\circ$.

① $\alpha > \theta$ $I(\alpha) = 0$ $\longrightarrow (1)$

② $\alpha < \theta$ $I(\alpha) = I(\pi+\alpha)$. $\longrightarrow (2)$.

① $\alpha = 30^\circ$. ; $\alpha < \theta$; $I(\alpha) = I(\pi+\alpha)$.

$\alpha = \omega t$; $t = \alpha/\omega$ ||. $\alpha' = \frac{30}{180} \times \pi \Rightarrow$ $\boxed{\pi/6. = \alpha}$

$I(\alpha) = 20.253 \sin(\alpha - 0.89863) + e^{-R/L \times \frac{\alpha}{\omega}} \times e^{c}$

$\boxed{I(\alpha) = 20.253 \sin(\pi/6 - 0.89863) + e^{-10/4\pi\,\alpha} \times e^{c}}$

$e^{-\frac{10}{4\pi}\alpha} \Rightarrow e^{-10/4\pi \times \pi/6} \Rightarrow 0.65924.$

$I(\alpha) = 20.253 \sin(\pi/6 - 0.89863) + 0.65924\, e^{c}$

$\boxed{I(\alpha) = -7.4187 + 0.65924\, e^{c}}$

$I(\pi+\alpha) = -20.253 \sin(\alpha - 0.89863) + e^{-10/4\pi \times (\pi+\alpha)} \times e^{c}$

$I(\pi+\alpha) = -20.253 \sin(\pi/6 - 0.89863) + e^{-10/4\pi \times (\pi+\pi/6)} \times e^{c}$.

$\boxed{I(\pi+\alpha) = 7.4187 + 0.0541\, e^{c}}$

$\sin(\pi+\alpha) = -\sin(\alpha)$

$\sin(-\theta) = -\sin(\theta)$.

$I(\alpha) = I(\pi + \alpha)$

$-7.4187 + 0.65924\, e^{c} = 7.4187 + 0.0541\, e^{c}$

$0.65924\, e^{c} = 14.8374 + 0.0541\, e^{c}$

$0.65924\, e^{c} - 0.0541\, e^{c} = 14.8374$

$e^{c}[0.65924 - 0.0541] = 14.8374$

$e^{c}[0.60514] = 14.8374$

$e^{c} = \dfrac{14.8374}{0.60514}$ $\boxed{e^{c} = 24.519\ A}$

$I_{total} = 20.253 \sin(200\pi t - 0.89863) + e^{-R/L} \times e^{c}$

$\boxed{I_{total} = 20.253 \sin(\alpha - 0.89863) + e^{-0.795 \gg \alpha} \times 24.519 A}$

$\alpha > \theta$; $60°, 90°$. $I(\alpha) = 0$

$$0 = 20.253 \sin(\alpha - 0.89863) + e^{-10/4\pi \alpha} \times e^{c} \quad \cdots (1)$$

$\alpha = \omega t$; $\alpha = 60°$, $\alpha = \frac{60}{180} \times \pi$ $\boxed{\alpha = \pi/3}$

$$0 = 20.253 \sin(\pi/3 - 0.89863) + e^{-10/4\pi \times \pi/3} \times e^{c}$$

$$0 = 2.99788 + 0.4346 \times e^{c}$$

$$\frac{-2.99788}{0.4346} = \boxed{e^{c} = -6.8980 \text{ A}}$$

$$I_{(\alpha)} = 20.253 \sin(\alpha - 0.89863) + e^{-10/4\pi \alpha} \times -6.8980 \text{ A}$$

$I_{(\beta)} \rightarrow$ extreme angle. At this the current will be zero!

$$\boxed{0 = 20.253 \sin(\beta - 0.89863) + e^{-10/4\pi \beta} - 6.8980 \text{ A}}$$

270, 90°, $I(\alpha) = 0$.

$$0 = 20.253 \sin(\alpha - 0.89863) + e^{-10/4\pi\alpha} \times e^{c}$$

$\alpha = 90°$; $\alpha = \frac{90}{180} \times \pi \Rightarrow \boxed{\alpha = \pi/2}$

$$0 = 20.253 \sin(\pi/2 - 0.89863) + e^{-10/4\pi \times \pi/2} \times e^{c}$$

$$0 = 12.611 + 0.2865\, e^{c}$$

$$\frac{-12.611}{0.2865} = \boxed{e^{c} = -44.017\,A}$$

$$\boxed{I(\alpha) = 20.253 \sin(\alpha - 0.89863) + e^{-10/4\pi\alpha} - 44.017\,A}$$

Extinction angle (β)

$$\boxed{0 = 20.253 \sin(\beta - 0.89863) + e^{-10/4\pi\beta} - 44.017\,A}$$

Average Voltage for $\alpha = 30° = \pi/6$

$\sin \to \cos \to -\sin \to -\cos$ (Diff)

$\sin \to -\cos \to -\sin \to \cos$ (Integ)

$$V_{avg} = \frac{V_m}{\pi} \int_{\alpha}^{\pi+2} \sin(\omega t)\, d(\omega t)$$

$$V_{avg} = \frac{325.27}{\pi} \int_{\pi/6}^{\pi/6+\pi} \sin(\omega t)\, d(\omega t)$$

$$V_{avg} = \frac{325.27}{\pi} \times \left[-\cos(\omega t) \right]_{\pi/6}^{\pi/6+\pi}$$

$$V_{avg} = \frac{325.27}{\pi} \times \left[-\cos(\pi/6 + \pi) + \cos(\pi/6) \right]$$

$$\boxed{V_{avg} = 179.33\,V}$$

$$\boxed{I_{avg} = \frac{V_{avg}}{R} = \frac{179.33}{10} \Rightarrow 17.933\,A}$$

For 60°, 90°, D.C.M:

$$\alpha = \frac{60}{180} \times \pi$$

$$\alpha = 90/180 \times \pi$$

$$V_{avg} = \frac{V_m}{\pi} \int_{\alpha}^{\beta} \sin(\omega t)\, d(\omega t)$$

$\beta \to$ Should be found easy matlab and ∴ can Calculate Average value.

For 30° Conduction:

Math: 179.33V

Matlab: 179.49V

For current: (30°)

Math: 17.933 A

Matlab: 17.726 A

Phase delay calculation:

Eg: 30° $T = 1/F = \frac{1}{100} \rightarrow (1)$ = Period.

Phase delay T_1, T_2 $= \frac{\alpha}{360} \times$ Period

$\hookrightarrow \frac{30}{360} \times \frac{1}{100} \rightarrow T_1, T_2.$

Phase delay T_3, T_4

$\hookrightarrow \frac{30 + \pi(180)}{360} \times \frac{1}{100} \Rightarrow T_3, T_4.$